PERMANENCE OF QUASI-COMPACTNESS AND QUASI-SEPARATEDNESS FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The three parts are handled inside a single proof environment; part (3) is the only one with a multi-step argument and it is written out fully.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Permanence of quasi-compactness in a triangle over the base. Lemma lemma-quasi-compact-permanence, used at proof lines 1488-1490. Gives part (1) with Z = S = Spec(Z).
- Composition statements for separatedness. Lemma lemma-compose-after-separated, used at proof lines 1491-1492. Gives the quasi-separated half of part (2).
- Stability of quasi-compactness and separatedness under base change. Lemmas lemma-base-change-quasi-compact and lemma-base-change-separated, used at proof lines 1493-1498. Makes X x_Y Z -> Z quasi-compact and quasi-separated.
- Quasi-compactness and quasi-separatedness of a space over a quasi-separated base. Lemmas lemma-separated-over-separated and lemma-composition-quasi-compact, used at proof lines 1499-1503. Concludes the two properties for the fibre product.

Source reference. Original lemma 03HF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4565–4565. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4681–4681. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4687–4687. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03HG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4566–4566. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04ZH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6465–6465. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of algebraic spaces over a scheme S.*

(1) *If X is quasi-compact and Y is quasi-separated, then f is quasi-compact.*
(2) *If X is quasi-compact and quasi-separated and Y is quasi-separated, then f is quasi-compact and quasi-separated.*
(3) *A fibre product of quasi-compact and quasi-separated algebraic spaces is quasi-compact and quasi-separated.*

Proof. Part (1) follows from Lemma 03KS with $Z = S = \mathrm{Spec}(\mathbf{Z})$. Part (2) follows from (1) and Lemma 03KR. For (3) let $X \to Y$ and $Z \to Y$ be morphisms of quasi-compact and quasi-separated algebraic spaces. Then $X \times_Y Z \to Z$ is quasi-compact and quasi-separated as a base change of $X \to Y$ using (2) and Lemmas 03HF and 03KL. Hence $X \times_Y Z$ is quasi-compact and quasi-separated as an algebraic space quasi-compact and quasi-separated over Z, see Lemmas 04ZH and 03HG. □